Zadanie 11 [4pkt] – TYP 3.3

Rozwiąż równanie: $(x^2 + 4x - 5)(2x^2 + 9x - 35) = 0$

$(x^2+4x-5)(2x^2+9x-35)=0$

$x^2+4x-5=0 \qquad 2x^2+9x-35=0$

$\Delta = b^2 - 4 \cdot a \cdot c$

$x_1 = \frac{-b-\sqrt{\Delta}}{2a} \qquad x_2 = \frac{-b+\sqrt{\Delta}}{2a}$

$\Delta_1 = 16+20=36 \qquad \Delta_2 = 81+280=361$

$\sqrt{\Delta_1} = 6 \qquad \sqrt{\Delta_2} = 19$

$x_1 = \frac{-4-6}{2} = -5 \qquad x_3 = \frac{-9-19}{4} = -7$

$x_2 = \frac{-4+6}{2} = 1 \qquad x_4 = \frac{-9+19}{4} = 2,5$

Odp: $x \in \{-7, -5, 1, 2\frac{1}{2}\}$.